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Peak Muse? Growth Slows Just As Morgan Stanley Tallies What Meta's "Free" Agent Really Costs

灵感见顶？增长放缓之际，摩根士丹利盘点 Meta“免费”智能体的真实成本

BY TYLER DURDEN

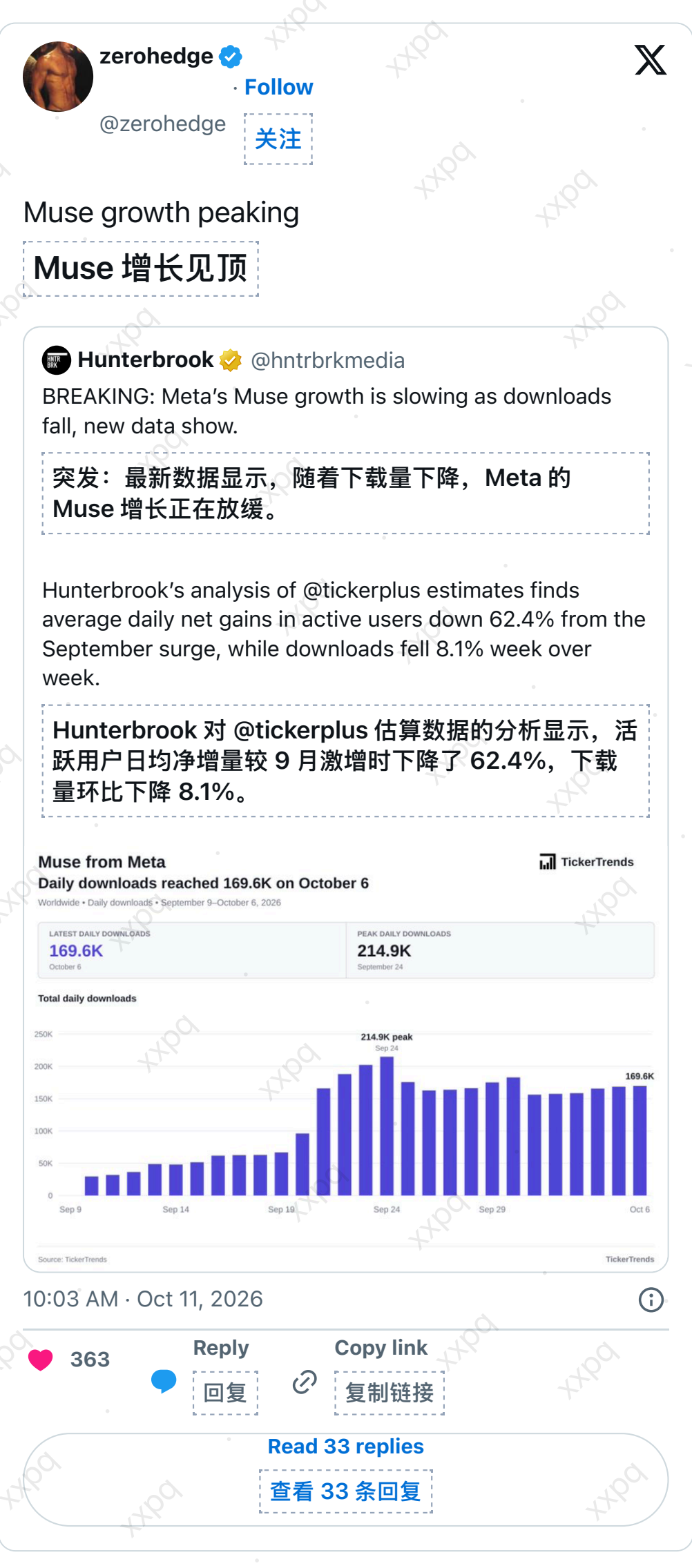
MONDAY, OCT 12, 2026 - 08:35 AM

For most of the past three years, the knock on Meta's AI strategy came down to one line: *lots of capex, no product*. Then just over a month ago, on September 8, Meta launched Muse, an agent that opens a browser, fills out forms, books the trip and (if you let it) cancels your gym membership and even reallocates your savings. Meta stock jumped 27% in 12 sessions to a $777.59 peak, the forward multiple went from 14x in late August to 21x, and the sell-side moved on from fretting about capex to arguing over which trillion-dollar TAM to stack on top of it.

在过去三年的大部分时间里，对 Meta 人工智能战略的诟病都归结为一句话：资本开支高企，却没有产品。随后，就在一个多月前的 9 月 8 日，Meta 推出了 Muse——一款能够打开浏览器、填写表格、预订行程的智能体；如果你允许，它还会取消你的健身房会员资格，甚至重新配置你的储蓄。Meta 股价在 12 个交易日内上涨 27%，触及 777.59 美元的峰值；远期市盈率从 8 月下旬的 14 倍升至 21 倍；卖方分析师也不再担心资本开支，转而争论该在此基础上叠加哪个万亿美元级的潜在市场。

It didn't take long, however, for the first serious crack to show up. On Friday, Hunterbrook Media citing alternative-data firm TickerTrends, **reported that Muse downloads fell 8.1% week over week and that daily net user additions have slowed 62.4% from the September surge.** There is, of course, a conflict of interest - Hunterbrook's affiliated fund disclosed it is short META - but that doesn't mean the data is wrong. We flagged it over the weekend with a three-word summary:

然而，首个明显裂缝很快便浮现。周五，Hunterbrook Media 援引另类数据公司 TickerTrends 的数据报道称，Muse 下载量环比下降 8.1%，每日净新增用户数较 9 月激增时放缓了 62.4%。当然，这里存在利益冲突——Hunterbrook 的关联基金披露其做空 META——但这并不意味着这些数据有误。周末，我们用三个词概括了这一情况：



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Muse growth peaking

Muse 增长见顶

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BREAKING: Meta's Muse growth is slowing as downloads fall, new data show.

突发：最新数据显示，随着下载量下降，Meta 的 Muse 增长正在放缓。

Hunterbrook's analysis of @tickerplus estimates finds average daily net gains in active users down 62.4% from the September surge, while downloads fell 8.1% week over week.

Hunterbrook 对 @tickerplus 估算数据的分析显示，活跃用户日均净增量较 9 月激增时下降了 62.4%，下载量环比下降 8.1%。

10:03 AM · Oct 11, 2026

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The timing is awkward for the bulls, because the most detailed piece of sell-side Muse work so far, Morgan Stanley's report "23 Days of Musing...Top Investor Questions and Our Answers" (*available to pro subs*), rests on a capacity analysis sized for 1 billion daily users. Brian Nowak, who has Meta as a Top Pick with a $775 target, built the first bottom-up Muse cost-to-serve model we've seen, along with a CPU capacity build, a Graviton deal sizing and a view on what Meta and Amazon are haggling over. Below we go through the slowdown data first, then all of Morgan Stanley's exhibits, with some napkin math the bank chose not to run and a look at what the slowdown does to each of its answers.

对多头而言，时机并不理想，因为迄今最详尽的卖方 Muse 研究——摩根士丹利报告《Muse 23 天……投资者最关心的问题及我们的回答》（Pro 订阅用户可查阅）——采用的是按每日 10 亿用户规模测算的容量分析。将 Meta 列为首选股、目标价定为 775 美元的 Brian Nowak，搭建了我们所见到的首个自下而上的 Muse 单位服务成本模型，同时还进行了 CPU 容量测算、Graviton 交易规模测算，并分析 Meta 与亚马逊正在就哪些问题讨价还价。下文先梳理增长放缓的数据，再逐一审视摩根士丹利的所有图表，并补上一些该行未做的粗略测算，最后看看增速放缓对其各项判断有何影响。

Peak Muse? What The Data Actually Show

Muse 热度见顶？数据实际显示了什么

Here is what TickerTrends' numbers, as analyzed by Hunterbrook, say:

以下是 Hunterbrook 分析 TickerTrends 数据后得出的结论：

- **Downloads:** ~169.6K on October 6, down 21.1% from the 214.9K peak on September 24, the day after Connect. Downloads have stayed below that peak for 12 straight days. On a weekly basis they averaged ~165K a day in the seven days to October 6, down 8.1% from ~180K the week before.

  下载量：10 月 6 日为 169.6K，较 9 月 24 日的峰值 214.9K 下降 21.1%。9 月 24 日是 Connect 发布后的第二天。此后连续 12 天，下载量均低于这一峰值。按周计算，截至 10 月 6 日的七天内，日均下载量为 165K，较前一周的 180K 下降 8.1%。

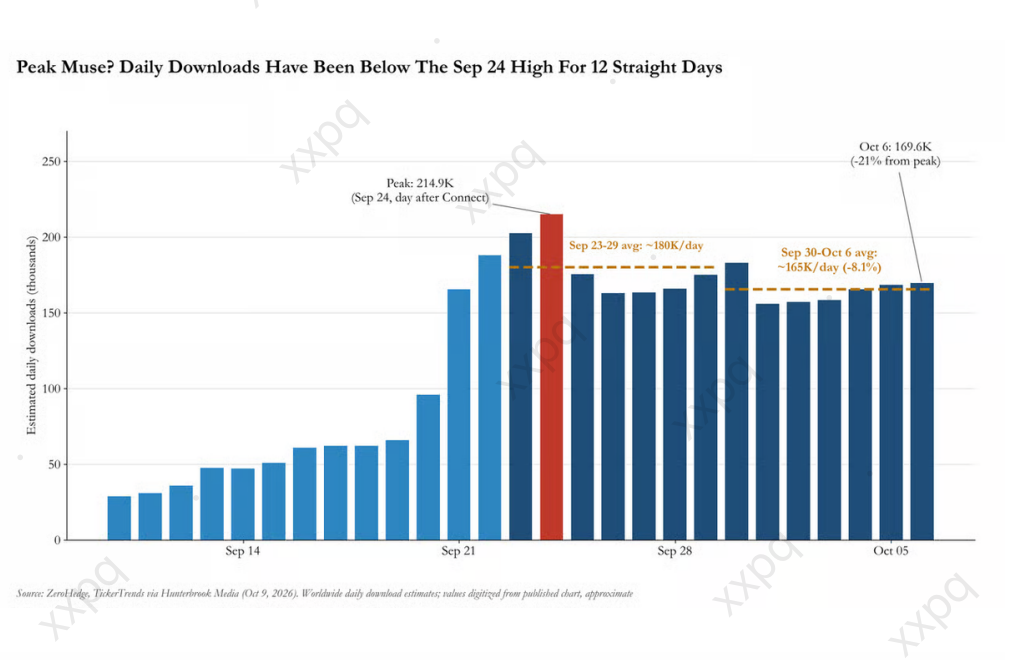
- **Users:** DAUs hit a new high of 2,730,919 on October 6, so the user base is still growing. The pace of growth is what changed. Net additions averaged ~198K a day over September 15-22, ~87K a day the following week and ~74K a day in the week to October 6, which is 62.4% below the initial surge and 14.5% below the prior week.

  用户：10 月 6 日，日活跃用户数（DAU）创下新高，达到 2,730,919，说明用户群仍在增长。变化的是增长速度。9 月 15 日至 22 日，日均净增用户数为 198K；随后一周为 87K；截至 10 月 6 日的一周为 74K，较最初的激增阶段低 62.4%，较前一周低 14.5%。

- **Engagement proxy:** app-review volume fell 11.6% week over week to 4,716 in the seven days through October 7, with iOS reviews down 22.5%. The daily count of 664 on October 7 was 17.2% below the October 1 peak of 802.

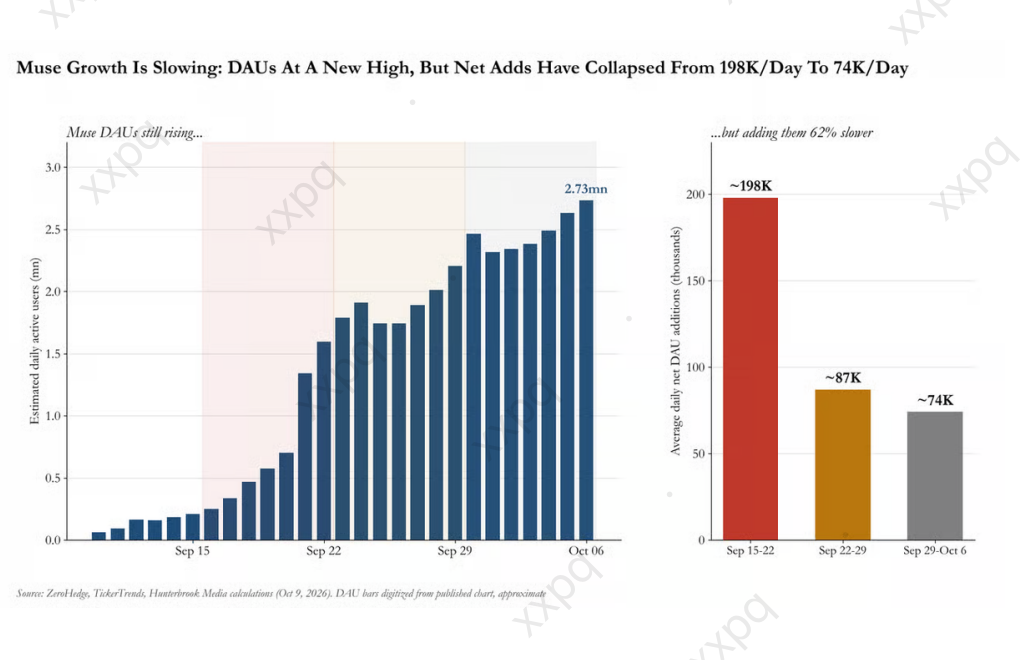
  参与度指标：截至 10 月 7 日的七天内，应用商店评论量环比下降 11.6%，至 4,716 条，其中 iOS 评论量下降 22.5%。10 月 7 日的单日评论量为 664 条，较 10 月 1 日 802 条的峰值低 17.2%。

Peak Muse? Daily Downloads Have Been Below The Sep 24 High For 12 Straight Days



Source: ZeroHedge, TickerTrends via Hunterbrook Media (Oct 9, 2026). Worldwide daily download estimates; values digitized from published chart, approximate

Muse Growth Is Slowing: DAUs At A New High, But Net Adds Have Collapsed From 198K/Day To 74K/Day



Source: ZeroHedge, TickerTrends, Hunterbrook Media calculations (Oct 9, 2026). DAU bars digitized from published chart, approximate

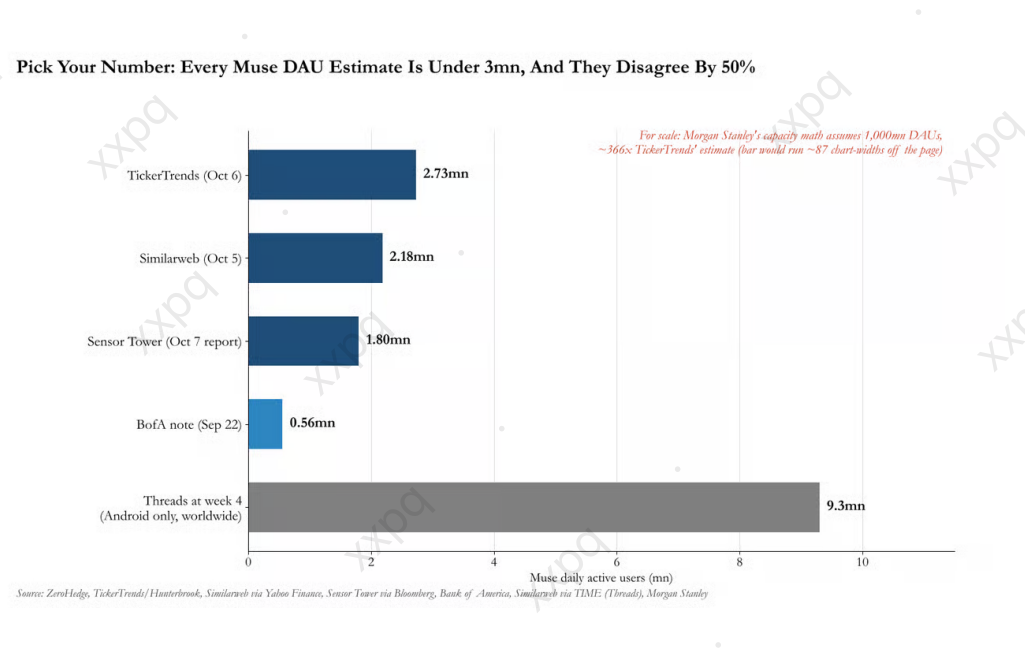
Hunterbrook also compared Muse with Threads, Meta's last viral launch, which was at ~9.3mn worldwide Android-only DAUs four weeks in (Similarweb), down from a peak near 50mn. Muse has fewer users than Threads had at the same point. In fairness, Threads was a free, global, sign-in-with-Instagram social app, while Muse launched in the US only (Canada followed September 18) and wants a payment card even on the free tier. Threads also bled users much faster after its debut, which Hunterbrook acknowledges.

Hunterbrook 还将 Muse 与 Meta 上一次引发病毒式传播的新产品 Threads 作了比较。上线四周时，Threads 全球 Android 端 DAU 为~9.3mn（Similarweb 数据），低于接近 50mn 的峰值。Muse 在同一时间点的用户数少于 Threads。公平地说，Threads 是一款免费、面向全球、可通过 Instagram 账号登录的社交应用；而 Muse 最初仅在美国推出（加拿大于 9 月 18 日跟进），且即使使用免费版也需要提供支付卡信息。Threads 在发布后用户流失速度也快得多，Hunterbrook 对此予以承认。

Two caveats before anyone declares Muse dead. First, Hunterbrook is short, and says so at the top of the piece. Second, nobody agrees on how many people use Muse. TickerTrends says 2.73mn DAUs. Similarweb, cited by Yahoo Finance on Thursday, had 2.18mn on October 5. Sensor Tower, per Bloomberg on October 7, had 6.6mn downloads and 1.8mn DAUs. That's a spread of about 50% between the highest and lowest estimate for the same week:

在有人宣布 Muse 已经无可救药之前，有两点需要注意。第一，Hunterbrook 持有空头头寸，并在文章开头明确说明了这一点。第二，没人对 Muse 的用户数达成共识。TickerTrends 称 DAU 为 2.73mn。雅虎财经周四援引的 Similarweb 数据显示，10 月 5 日 DAU 为 2.18mn。彭博社 10 月 7 日援引 Sensor Tower 称，下载量为 6.6mn，DAU 为 1.8mn。同一周内，最高和最低估值之间相差约 50%：

Pick Your Number: Every Muse DAU Estimate Is Under 3mn, And They Disagree By 50%



Source: ZeroHedge, TickerTrends/Hunterbrook, Similarweb via Yahoo Finance, Sensor Tower via Bloomberg, Bank of America, Similarweb via TIME (Threads), Morgan Stanley

Every one of those estimates is under 3mn, and Morgan Stanley's capacity analysis below is built around 1,000mn. On TickerTrends' number, Meta has already bought CPUs for roughly 366 times the users it has. To be fair, the note says plainly that it chose an aggressive adoption case "to show the large amount of capacity Meta already has coming on." But it does put the "Muse is taking over the internet" commentary in perspective.

上述所有估值均低于 3mn，而摩根士丹利下文的产能分析则以 1,000mn 为基础。按 TickerTrends 的数据，Meta 已经采购的 CPU，足以服务其现有用户数的约 366 倍。公平地说，报告明确表示，采用激进的普及情景是“为了展示 Meta 即将获得的庞大产能”。但这确实让“Muse 正在接管互联网”这一说法显得更有分寸。

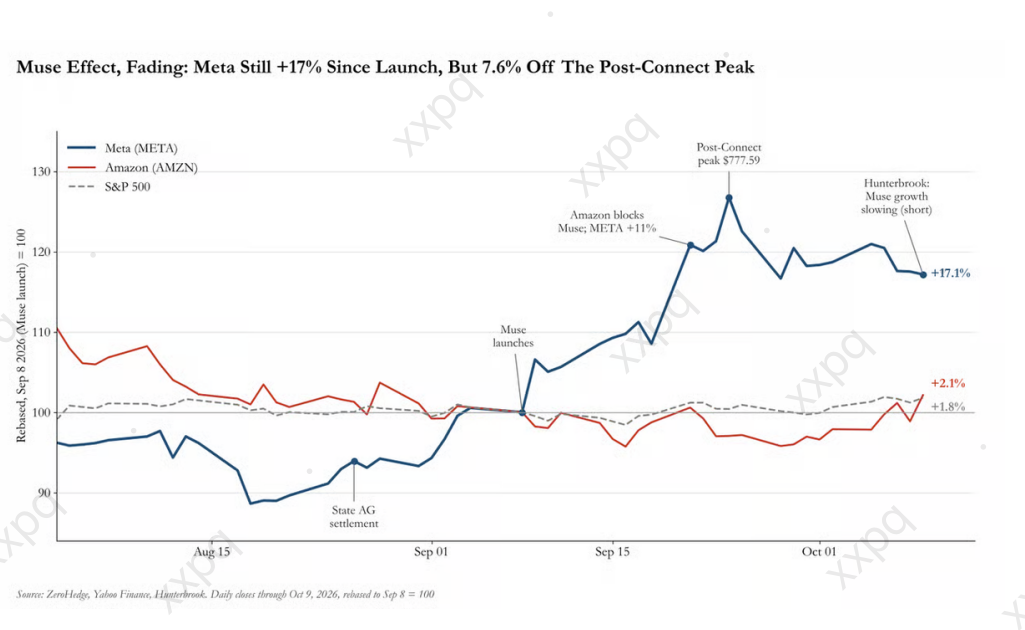
Not everyone on the sell-side is fazed. Deutsche Bank's Benjamin Black, writing on the Similarweb data, said Meta has "found a clear gap amongst current AI-agentic offerings" and sees Muse reaching up to 8% of Meta's revenue by 2030 (a range of ~$2.4bn to ~$36bn). JPMorgan's Doug Anmuth, who raised his target to $920 after Connect, has stressed that "Meta is not focused on the monetization of Muse in the near term," which is a polite way of saying investors are paying for growth for now. That makes a growth slowdown more relevant to the stock than it would be for a product already throwing off cash.

并非所有卖方分析师都对此感到不安。德意志银行的本杰明·布莱克在分析 Similarweb 数据时表示，Meta“在当前的 AI 智能体产品中发现了一个明显的市场空白”，并预计到 2030 年，Muse 的收入最高可达 Meta 总营收的 8%（区间为 ~24 亿美元至 ~360 亿美元）。Connect 大会后将目标价上调至 920 美元的摩根大通分析师道格·安穆斯强调，“Meta 短期内并未专注于 Muse 的变现”。委婉地说，这意味着投资者目前买单的是增长。因此，与已经开始创造现金流的产品相比，增长放缓对 Meta 股价的影响更大。

The market's verdict so far: a shrug. META slipped 0.3% on Friday to $718.67 after the report. It's still up 17.1% since launch but 7.6% below its September 24 high. Over the same span the S&P is up 1.8%, and Amazon, which blocked Muse from its store, has recovered to +2.1%.

迄今为止，市场的反应是：不以为意。报告发布后，META 周五下跌 0.3%，收于 718.67 美元。自上线以来，该股仍上涨 17.1%，但较 9 月 24 日的高点低 7.6%。同期，标普 500 指数上涨 1.8%；曾将 Muse 拒之门外的亚马逊则已反弹 2.1%。

Muse Effect, Fading: Meta Still +17% Since Launch, But 7.6% Off The Post-Connect Peak



Source: ZeroHedge, Yahoo Finance, Hunterbrook. Daily closes through Oct 9, 2026; rebased to Sep 8 = 100

How We Got Here

A quick recap. Meta introduced Muse on September 8. It runs on the Muse Spark model, gives each user a dedicated "Muse Secure VM", and lives in its own app, on the web, inside WhatsApp and on a native Mac client. Pricing works like a buffet: a free tier with a 100mn-token weekly allowance, a $20/month "Power" tier (500mn tokens a week) and a $100/month "Maximum" tier (3bn tokens a week). Connect (September 23) added a realtime avatar, Mac computer control, glasses integration and a palm-sized 5G Muse device, and Muse for Small Business followed September 29. On iOS Muse logged 1.43mn downloads in its first 12 days vs. ChatGPT's 1.37mn over the same window, per JPMorgan, and hit No. 1 on both the App Store and Google Play.

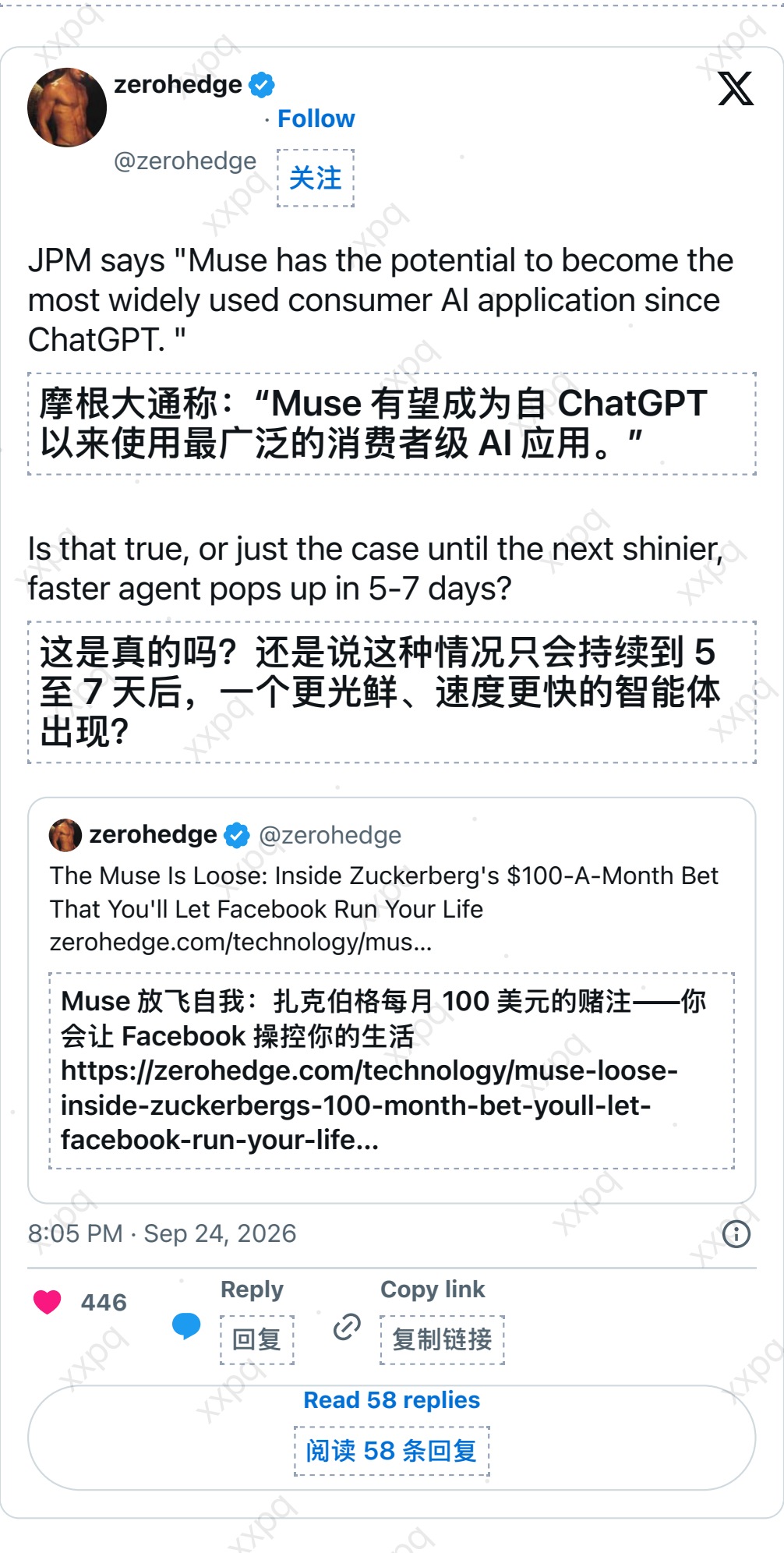
简要回顾。Meta 于 9 月 8 日推出 Muse。它基于 Muse Spark 模型运行，为每位用户提供专属的“Muse Secure VM”，并以独立应用、网页版、WhatsApp 内置服务和原生 Mac 客户端等形式提供。其定价类似自助餐：免费档每周提供 100mn 个 token；每月 20 美元的“Power”档每周提供 500mn 个 token；每月 100 美元的“Maximum”档每周提供 3bn 个 token。9 月 23 日推出的 Connect 增加了实时虚拟形象、Mac 电脑控制、眼镜集成，以及一款手掌大小的 5G Muse 设备；面向小型企业的 Muse 则于 9 月 29 日推出。据摩根大通数据，Muse 在 iOS 上线后的前 12 天下载量达 143 万次，同期 ChatGPT 为 137 万次；Muse 还登上了 App Store 和 Google Play 的榜首。

Then the friction started. Amazon blocked Muse from its site on the evening of September 20. TickerTrends now tracks user reports of restrictions at Costco and Netflix as well, and telecom carriers have reportedly started routing Muse's AI-originated customer-service calls away from human agents, which neuters the "call on my behalf" feature. A zero-day that could let a local app hijack a user's Muse account had to be hot-fixed. Hunterbrook separately reported last month that Muse could be prompted to compile dossiers on members of vulnerable groups. Meta also conceded that Muse's workspace file structure followed the OpenClaw blueprint, though it says the code was written from scratch.

随后，摩擦开始显现。亚马逊于 9 月 20 日晚间屏蔽了 Muse 对其网站的访问。TickerTrends 目前还在跟踪用户报告称，Costco 和 Netflix 也对 Muse 实施了限制；据报道，电信运营商已开始将由 Muse 发起的 AI 客服电话转接至人工客服之外的渠道，致使“代我致电”功能形同虚设。一个可能导致本地应用劫持用户 Muse 账户的零日漏洞不得不通过热修复解决。Hunterbrook 上月另行报道称，用户可以通过提示 Muse 为弱势群体成员整理档案。Meta 还承认，Muse 的工作区文件结构沿用了 OpenClaw 的蓝图，不过称相关代码是从头编写的。

Regular readers will recall we covered the launch and the Amazon block in "Agentic Wars Begin: Amazon Blocks Meta's Agents As Muse Takes World By Storm", followed by the Connect deep-dive "The Muse Is Loose: Inside Zuckerberg's $100-A-Month Bet That You'll Let Facebook Run Your Life". And when JPMorgan declared at the peak of the hype that Muse "has the potential to become the most widely used consumer AI application since ChatGPT," we asked the obvious follow-up:

细心的读者会记得，我们曾在《智能体之战打响：亚马逊封杀 Meta 智能体，Muse 席卷全球》中报道 Muse 的推出以及亚马逊的封锁，随后又在深度报道《Muse 脱缰：扎克伯格每月押注 100 美元，只为让你把生活交给 Facebook 打理》中深入探讨了 Connect。JPMorgan 在热潮巅峰时宣称，Muse“有潜力成为自 ChatGPT 以来使用最广泛的消费级 AI 应用”，我们当时便追问了一个显而易见的问题：



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JPM says "Muse has the potential to become the most widely used consumer AI application since ChatGPT. "

摩根大通称：“Muse 有望成为自 ChatGPT 以来使用最广泛的消费者级 AI 应用。”

Is that true, or just the case until the next shinier, faster agent pops up in 5-7 days?

这是真的吗？还是说这种情况只会持续到 5 至 7 天后，一个更光鲜、速度更快的智能体出现？

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8:05 PM · Sep 24, 2026

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Downloads peaked that same day. We'll allow ourselves a modest victory lap and move on to Morgan Stanley's six questions.

1. "How And When Could Meta Monetize Muse?" Answer: Not Soon

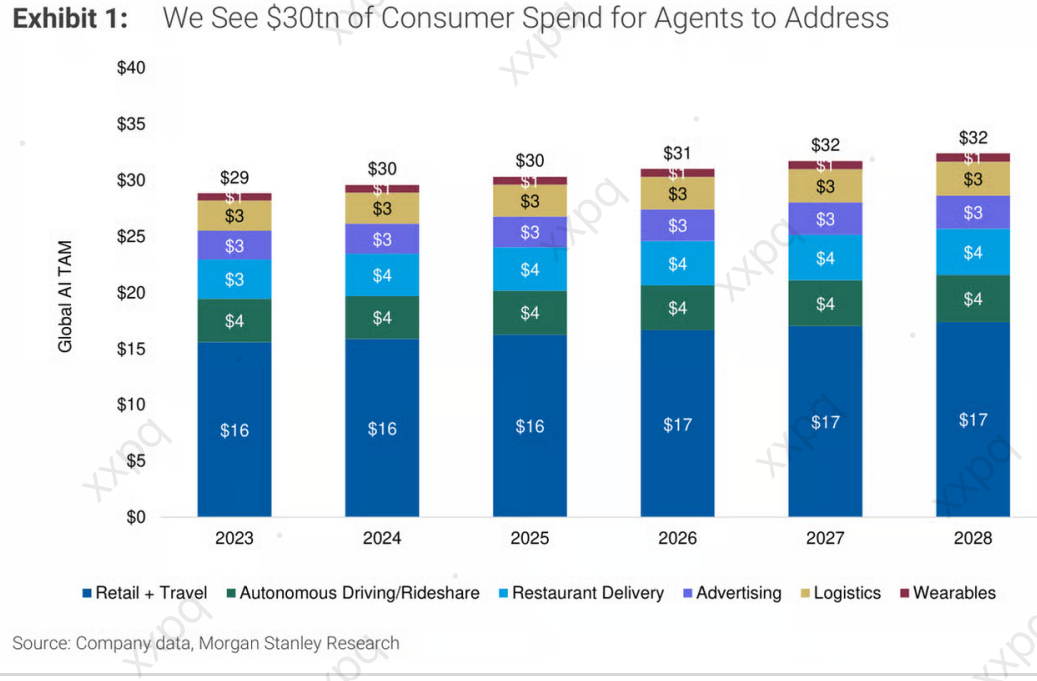
The honest answer comes first, and it's refreshingly blunt for a Top Pick note:

> "First, we expect Muse subscription revenue to be quite small for a while as Meta focuses on scaling a user base, engagement and commercial behavior well before monetization. To this point, we believe initial 'take rates' on the connection agreements are de minimis. **But over the long-term as the agentic behavior scales (think late '27/'28 at the earliest) we would expect Meta to monetize through a real-time bidding auction market (businesses paying for transactions/acquisitions)** with businesses across the economy bidding based on return on investment and paying Meta for completed transactions." (emphasis ours)

Translation: Meta wants to build for agents the same thing it built for ads, an auction where businesses bid for outcomes. This time the outcome is a completed purchase made by your agent. Nowak thinks the pool this eventually addresses is about $30 trillion of consumer spend, of which roughly $17 trillion is retail and travel:

Meta 希望为智能体打造与其广告业务相同的东西：一个由企业竞价争取转化结果的拍卖机制。这一次，转化结果是由你的智能体完成的一笔购买。Nowak 认为，这最终所针对的消费支出规模约为 30 万亿美元，其中零售和旅游约占 17 万亿美元：

Exhibit 1: We See $30tn of Consumer Spend for Agents to Address



Source: Company data, Morgan Stanley Research

Source: Morgan Stanley 来源：摩根士丹利

Morgan Stanley isn't alone on timing. BofA, which reiterated Buy with an $810 target on September 22, said material revenue from ads, subscriptions or commissions is unlikely before 2028. Goldman's Eric Sheridan, who reiterated Buy on September 13 and called Meta's "scale and distribution" the "most difficult-to-replicate assets in the consumer AI agent competition," reportedly framed the "key question" as "whether monetization swiftly follows," and flagged privacy and security as the single biggest risk. A week later came the zero-day. Goldman's $725 target, for the record, was overtaken by the stock eight days after it was set. Price targets tend to follow price.

对于这一时间判断，摩根士丹利并非孤例。美国银行（BofA）于 9 月 22 日重申“买入”评级，目标价为 810 美元，并表示广告、订阅或佣金带来的实质性收入不太可能在 2028 年前出现。高盛的 Eric Sheridan 于 9 月 13 日重申“买入”评级，并称 Meta 的“规模与分发能力”是消费级 AI 智能体竞争中“最难以复制的资产”。据报道，他将“关键问题”概括为“变现能否迅速跟进”，并指出隐私和安全是最大的单一风险。一周后，零日漏洞事件发生。顺带一提，高盛给出的 725 美元目标价，在设定八天后就被股价超越。目标价往往追随股价。

The slowdown matters most here. An auction for agent-driven transactions needs a very large, very active user base before businesses will bid real money for it. Every week that net adds run at 74K a day instead of 198K pushes that "late '27/'28 at the earliest" further out. In the meantime MS says Muse is "a further source of data capture and profile build for Meta's core products." Until the agent pays for itself, it pays by learning what you want, which happens to be how Meta's existing business makes money.

在这里，增长放缓尤为重要。智能体驱动交易的拍卖机制，需要先拥有庞大且活跃的用户群体，企业才会愿意为此投入真金白银竞价。日均净增用户数从 19.8 万降至 7.4 万，每持续一周，“最早要到 2027 年末/2028 年”的时间预期就会进一步推迟。与此同时，摩根士丹利称，Muse“进一步为 Meta 核心产品的数据采集和用户画像构建提供了来源”。在智能体尚未实现自我变现之前，它会通过了解你的需求来创造价值——而这恰好也是 Meta 现有业务的赚钱方式。

2. "How Much Are Muse Serving Costs?" Anywhere From $3 To $130 A Month. Per User.

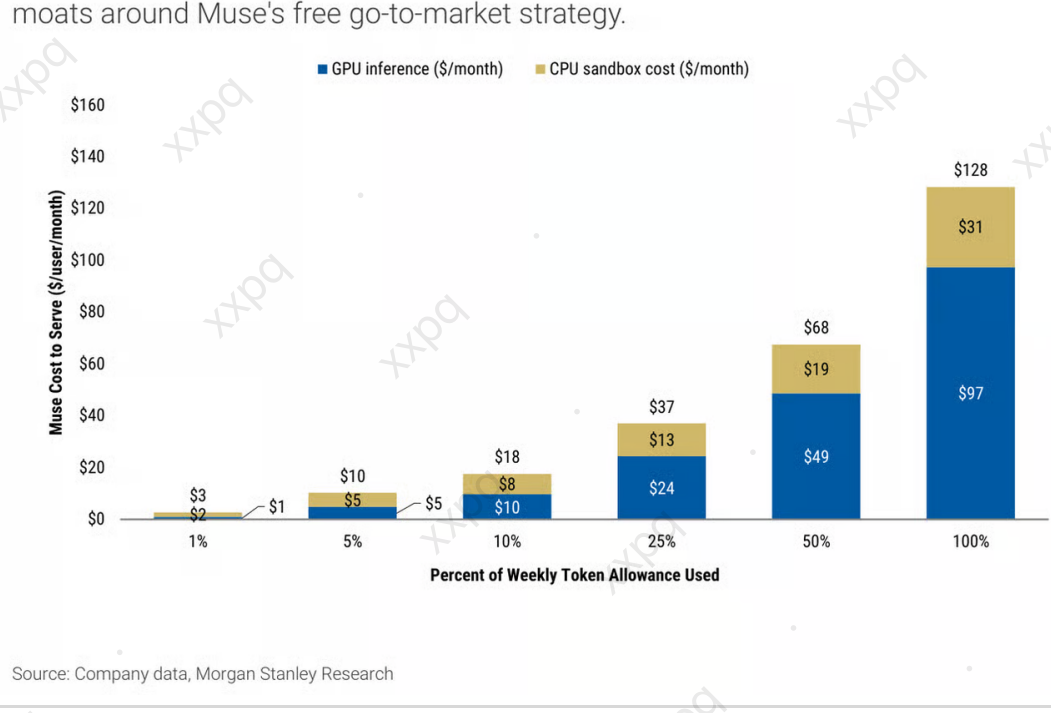
2. “Muse 的推理服务成本是多少？”每位用户每月 3 至 130 美元不等。

This is the main event. Morgan Stanley splits Muse's compute bill into two pieces: **GPU inference**, which scales with tokens consumed (the model reasoning through your request), and the **CPU sandbox**, which scales with how often you activate Muse and how long your VM stays provisioned (the browser, tools and code execution). The result:

这是核心议题。摩根士丹利将 Muse 的计算成本拆分为两部分：GPU 推理成本，随消耗的 token 数量而变化（即模型针对你的请求进行推理）；以及 CPU 沙盒成本，随 Muse 的启动频率和虚拟机保持配置状态的时长而变化（包括浏览器、工具和代码执行）。结果如下：

> "Our sensitivity work showcases how **it ranges from $3-$130 per user per month based on token consumption**... The mix of these costs shifts with intensity as the CPU sandbox expense is the larger cost driver for lower-usage users, while GPU inference becomes the dominant component for users consuming the full token allowance." (emphasis ours)
>
> “我们的敏感性分析显示，按 token 消耗量计算，每位用户每月的成本在 3 至 130 美元之间……随着使用强度增加，各项成本的构成也会变化：对低使用量用户而言，CPU 沙盒费用是更大的成本驱动因素；而对消耗全部 token 配额的用户而言，GPU 推理则成为主要成本组成部分。”（着重号为我们所加）

Exhibit 2: We estimate Muse cost to serve per user is anywhere from $3-$130/month...depending on token consumption. This speaks to the cost prohibitive moats around Muse's free go-to-market strategy.



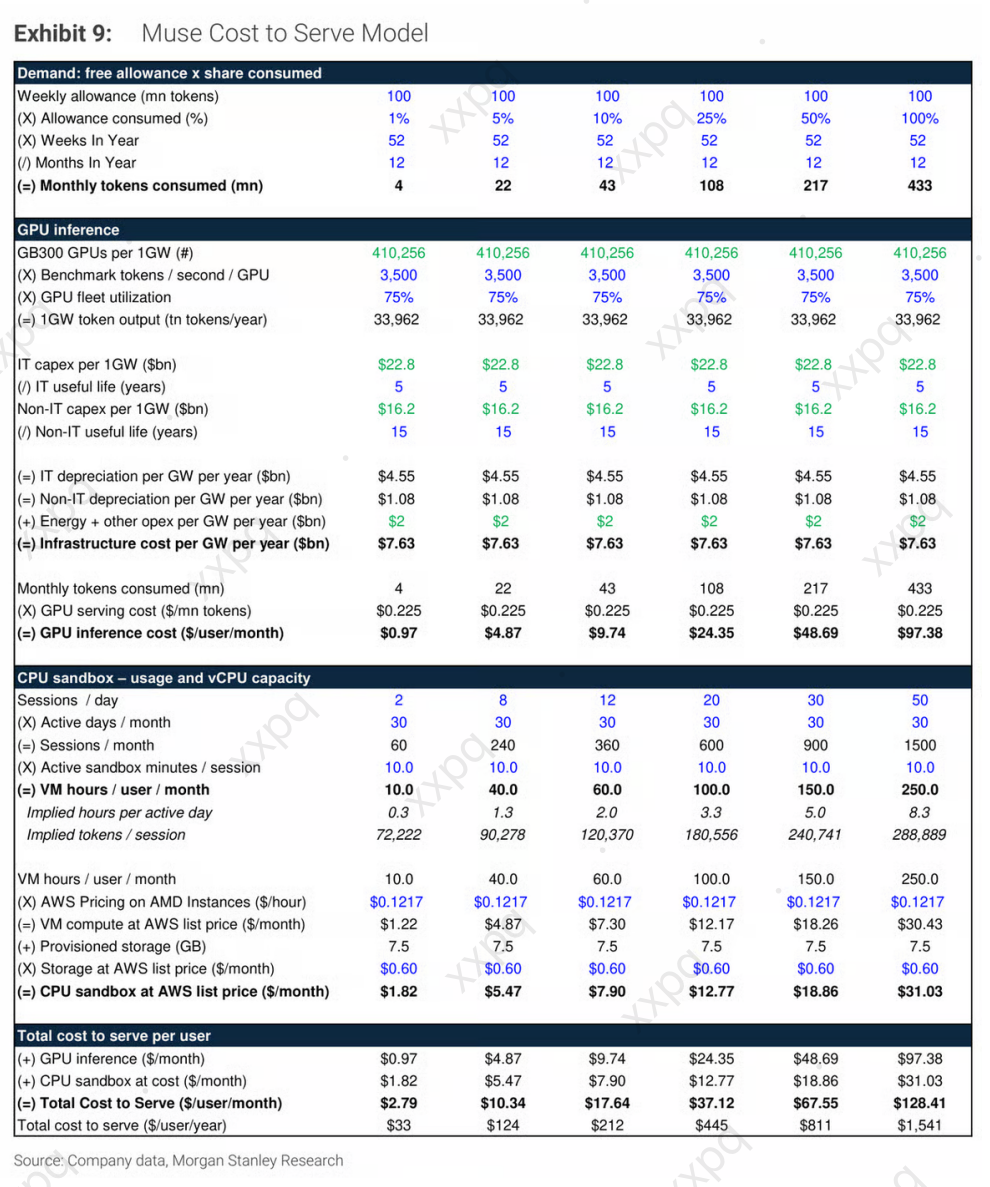
Source: Company data, Morgan Stanley Research

Source: Morgan Stanley 来源：摩根士丹利

The full model is worth studying line by line, because every assumption is in plain sight. On the GPU side, MS uses its GB300 framework from *The Paths to 25-50% GenAI ROIC* (July 27): 410,256 GPUs per gigawatt, 3,500 tokens/second/GPU at 75% utilization, which works out to 33,962 trillion tokens per GW per year. Against $7.63bn of annual infrastructure cost per GW (IT depreciation of $4.55bn, non-IT depreciation of $1.08bn and $2bn of energy and other opex), that gives a serving cost of **$0.225 per million tokens**. On the CPU side, MS assumes ~10 minutes of provisioned sandbox per activation, a 2 vCPU/8GB VM priced at AWS on-demand rates for AMD instances ($0.1217/hour), and 7.5GB of storage at $0.60/month.

值得逐行研究完整模型，因为每项假设都明明白白地列了出来。在 GPU 方面，摩根士丹利采用了其《通往 25%–50% 生成式 AI 投资资本回报率之路》（7 月 27 日）中的 GB300 框架：每吉瓦配备 410,256 块 GPU；在 75% 利用率下，每块 GPU 每秒处理 3,500 个 token，据此计算，每吉瓦每年可处理 33,962 万亿个 token。按每吉瓦每年 76.3 亿美元的基础设施成本计算（IT 折旧 45.5 亿美元、非 IT 折旧 10.8 亿美元，以及 20 亿美元的能源和其他运营支出），推算出的推理服务成本为每百万个 token 0.225 美元。在 CPU 方面，摩根士丹利假设每次启动配置沙盒 ~10 分钟，使用一台 2 vCPU/8GB 的虚拟机，按 AWS AMD 实例的按需费率定价（每小时 0.1217 美元），另有 7.5GB 存储，每月 0.60 美元。

Exhibit 9: Muse Cost to Serve Model



Demand: free allowance x share consumed						
Weekly allowance (mn tokens)	100	100	100	100	100	100
(X) Allowance consumed (%)	1%	5%	10%	25%	50%	100%
(X) Weeks In Year	52	52	52	52	52	52
(/) Months In Year	12	12	12	12	12	12
(=) Monthly tokens consumed (mn)	**4**	**22**	**43**	**108**	**217**	**433**
GPU inference						
GB300 GPUs per 1GW (#)	410,256	410,256	410,256	410,256	410,256	410,256
(X) Benchmark tokens / second / GPU	3,500	3,500	3,500	3,500	3,500	3,500
(X) GPU fleet utilization	75%	75%	75%	75%	75%	75%
(=) 1GW token output (tn tokens/year)	33,962	33,962	33,962	33,962	33,962	33,962
IT capex per 1GW ($bn)	$22.8	$22.8	$22.8	$22.8	$22.8	$22.8
(/) IT useful life (years)	5	5	5	5	5	5
Non-IT capex per 1GW ($bn)	$16.2	$16.2	$16.2	$16.2	$16.2	$16.2
(/) Non-IT useful life (years)	15	15	15	15	15	15
(=) IT depreciation per GW per year ($bn)	$4.55	$4.55	$4.55	$4.55	$4.55	$4.55
(=) Non-IT depreciation per GW per year ($bn)	$1.08	$1.08	$1.08	$1.08	$1.08	$1.08
(+) Energy + other opex per GW per year ($bn)	$2	$2	$2	$2	$2	$2
(=) Infrastructure cost per GW per year ($bn)	**$7.63**	**$7.63**	**$7.63**	**$7.63**	**$7.63**	**$7.63**
Monthly tokens consumed (mn)	4	22	43	108	217	433
(X) GPU serving cost ($/mn tokens)	$0.225	$0.225	$0.225	$0.225	$0.225	$0.225
(=) GPU inference cost ($/user/month)	**$0.97**	**$4.87**	**$9.74**	**$24.35**	**$48.69**	**$97.38**
CPU sandbox – usage and vCPU capacity						
Sessions / day	2	8	12	20	30	50
(X) Active days / month	30	30	30	30	30	30
(=) Sessions / month	60	240	360	600	900	1500
(X) Active sandbox minutes / session	10.0	10.0	10.0	10.0	10.0	10.0
(=) VM hours / user / month	**10.0**	**40.0**	**60.0**	**100.0**	**150.0**	**250.0**
Implied hours per active day	*0.3*	*1.3*	*2.0*	*3.3*	*5.0*	*8.3*
Implied tokens / session	*72,222*	*90,278*	*120,370*	*180,556*	*240,741*	*288,889*
VM hours / user / month	10.0	40.0	60.0	100.0	150.0	250.0
(X) AWS Pricing on AMD Instances ($/hour)	$0.1217	$0.1217	$0.1217	$0.1217	$0.1217	$0.1217
(=) VM compute at AWS list price ($/month)	$1.22	$4.87	$7.30	$12.17	$18.26	$30.43
(+) Provisioned storage (GB)	7.5	7.5	7.5	7.5	7.5	7.5
(X) Storage at AWS list price ($/month)	$0.60	$0.60	$0.60	$0.60	$0.60	$0.60
(=) CPU sandbox at AWS list price ($/month)	**$1.82**	**$5.47**	**$7.90**	**$12.77**	**$18.86**	**$31.03**
Total cost to serve per user						
(+) GPU inference ($/month)	$0.97	$4.87	$9.74	$24.35	$48.69	$97.38
(+) CPU sandbox at cost ($/month)	$1.82	$5.47	$7.90	$12.77	$18.86	$31.03
(=) Total Cost to Serve ($/user/month)	**$2.79**	**$10.34**	**$17.64**	**$37.12**	**$67.55**	**$128.41**
Total cost to serve ($/user/year)	$33	$124	$212	$445	$811	$1,541

Source: Company data, Morgan Stanley Research